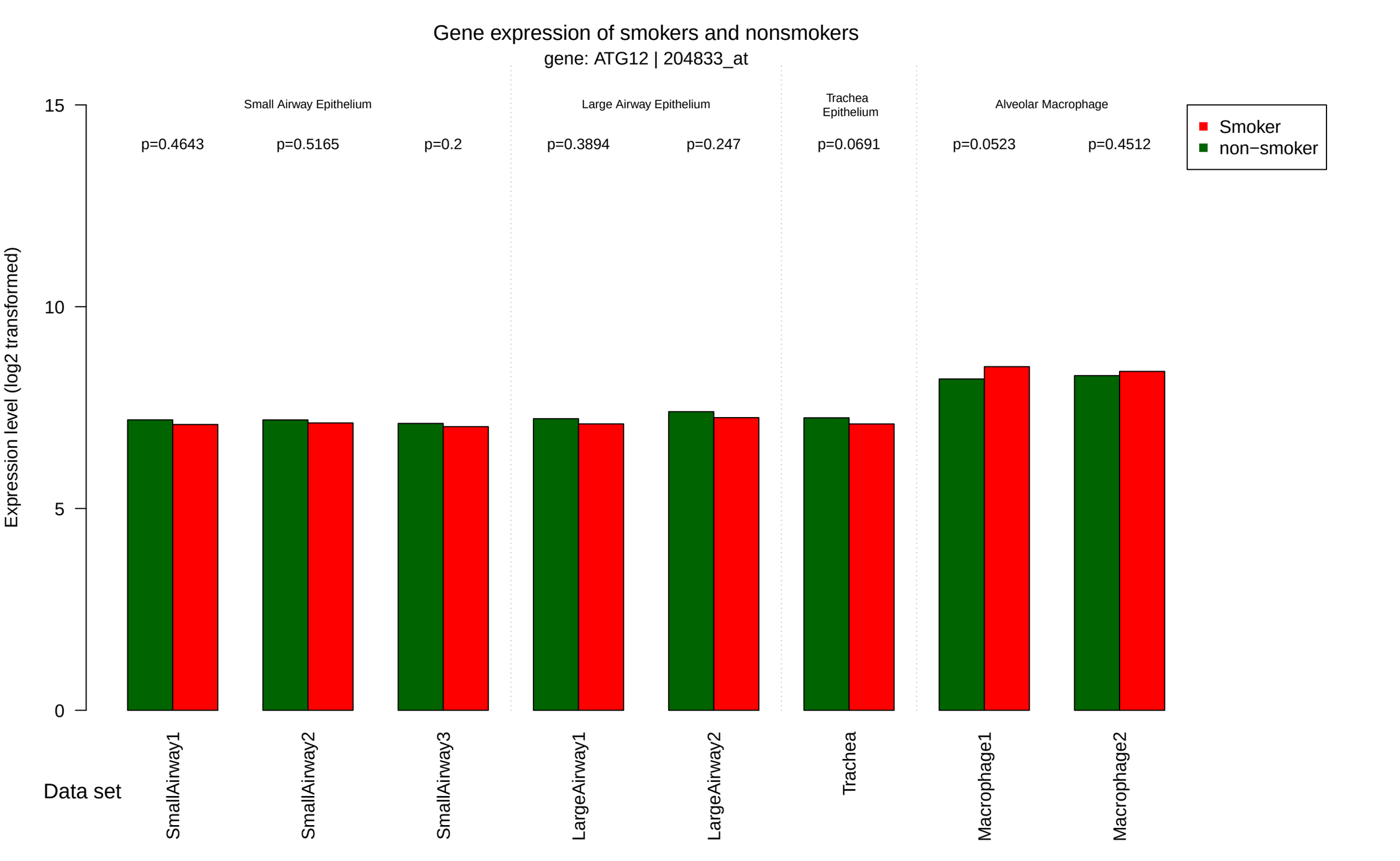
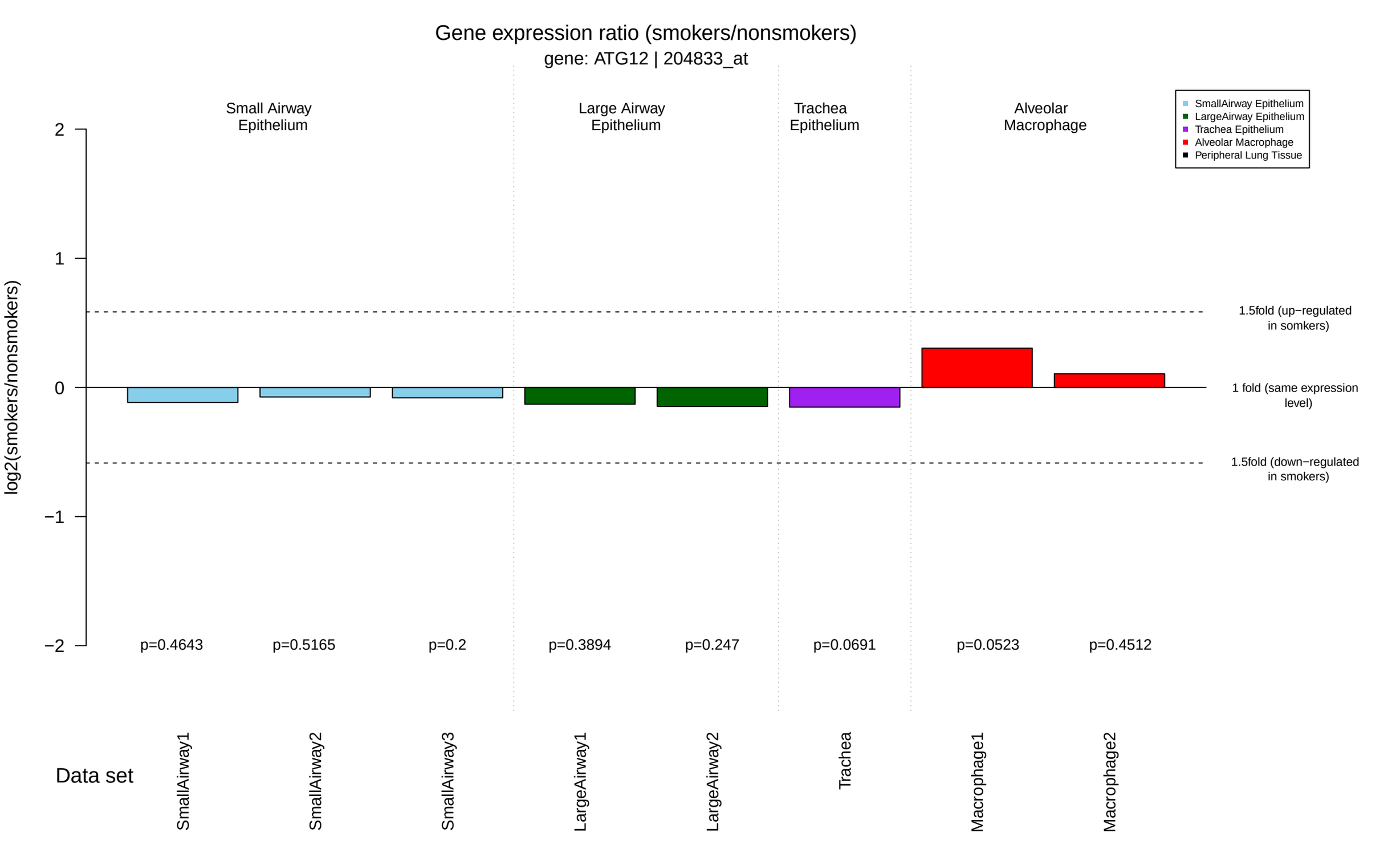
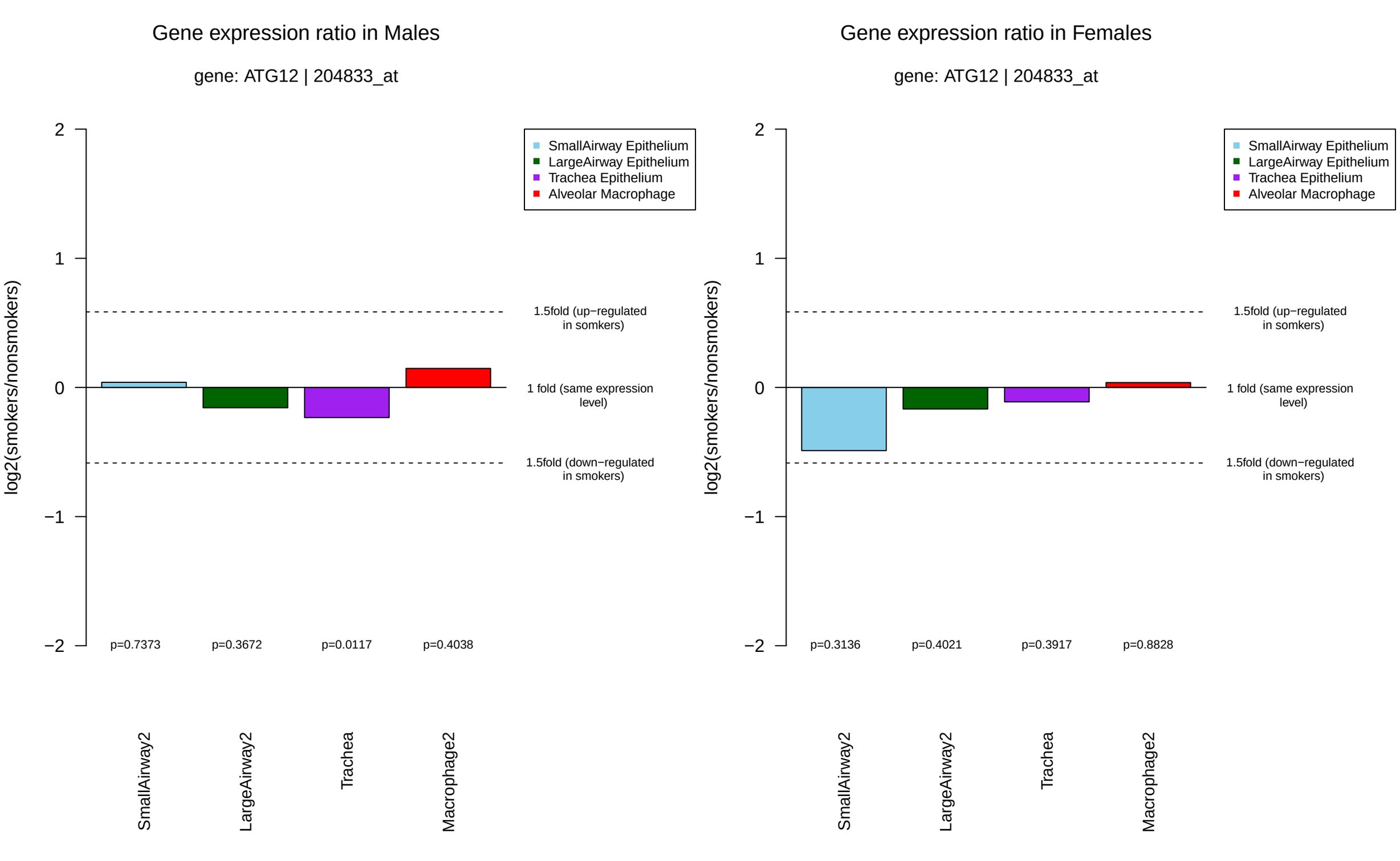
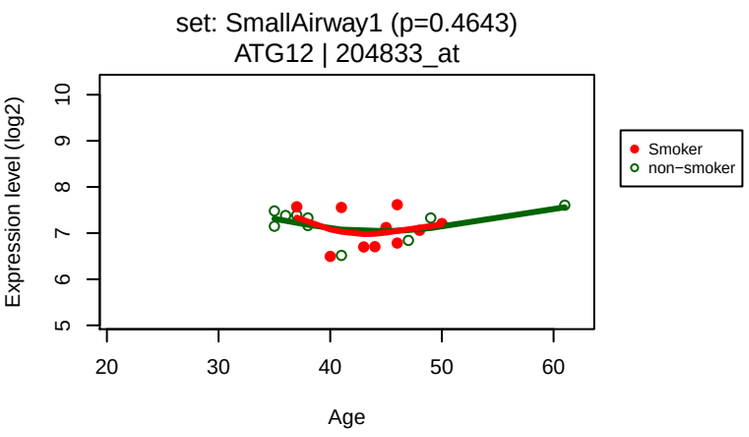
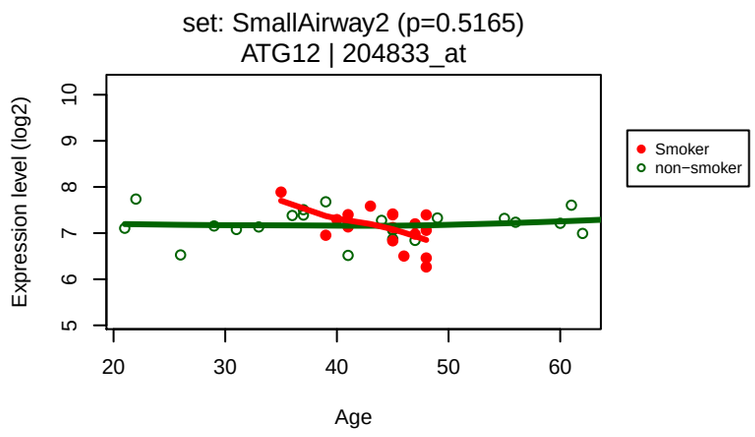
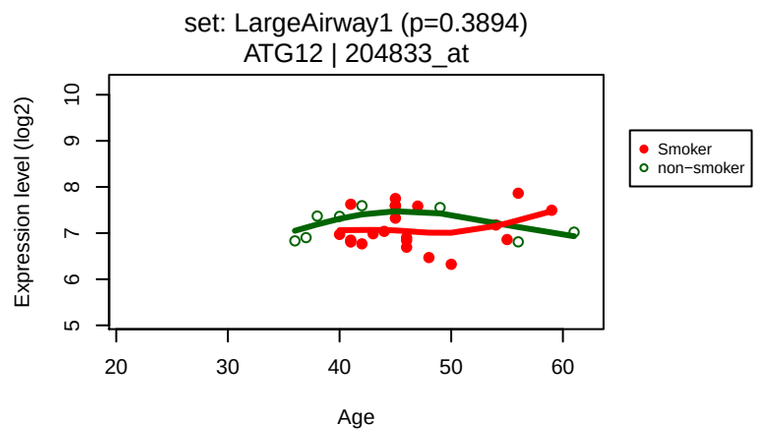

Gene expression of smokers and nonsmokers
gene: ATG12 | 204833_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4643
p=0.5165
p=0.2
p=0.3894
p=0.247
p=0.0691
p=0.0523
p=0.4512
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

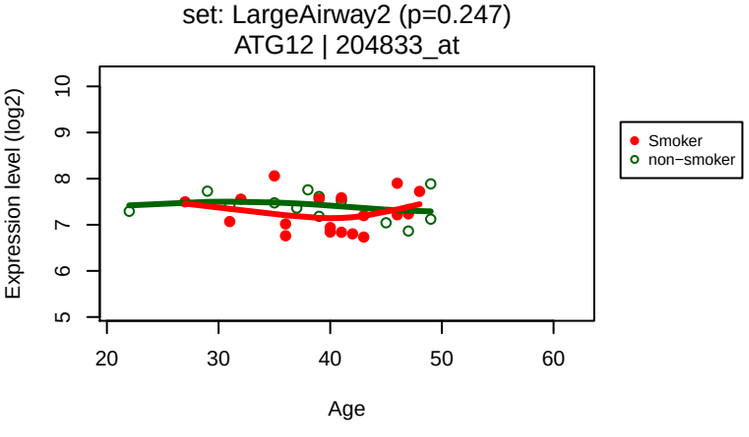

Gene expression ratio (smokers/nonsmokers)
gene: ATG12 | 204833_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4643
p=0.5165
p=0.2
p=0.3894
p=0.247
p=0.0691
p=0.0523
p=0.4512
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

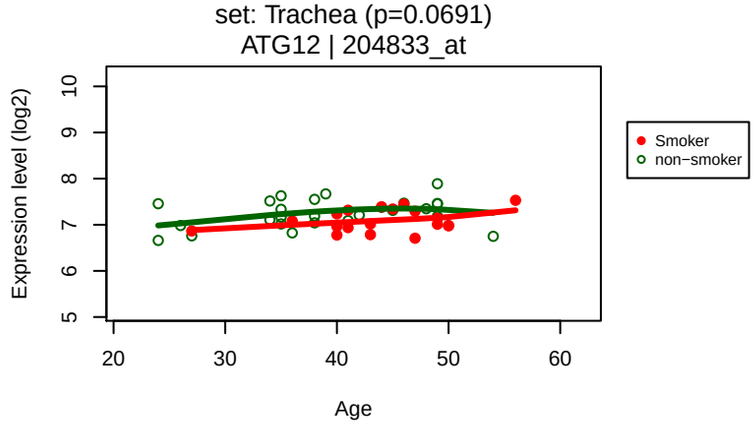

Gene expression ratio in Males
gene: ATG12 | 204833_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7373
p=0.3672
p=0.0117
p=0.4038
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATG12 | 204833_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3136
p=0.4021
p=0.3917
p=0.8828
SmallAirway2
LargeAirway2
Trachea
Macrophage2

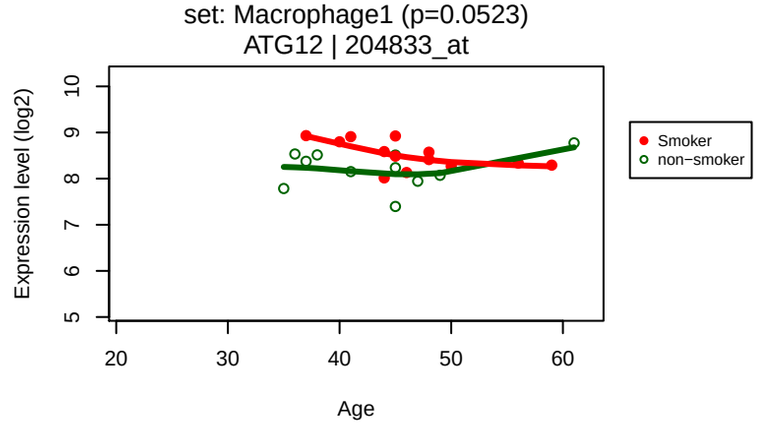

set: SmallAirway1 (p=0.4643)
ATG12 | 204833_at
Expression level (log2)
Age
Smoker
non−smoker

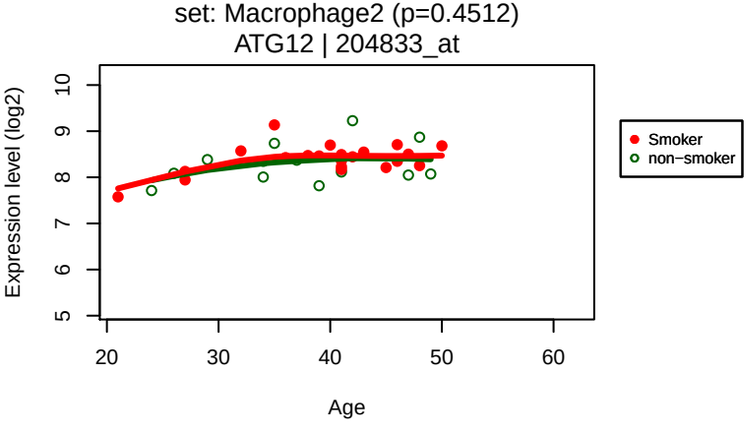

set: SmallAirway2 (p=0.5165)
ATG12 | 204833_at
Expression level (log2)
Age
Smoker
non−smoker

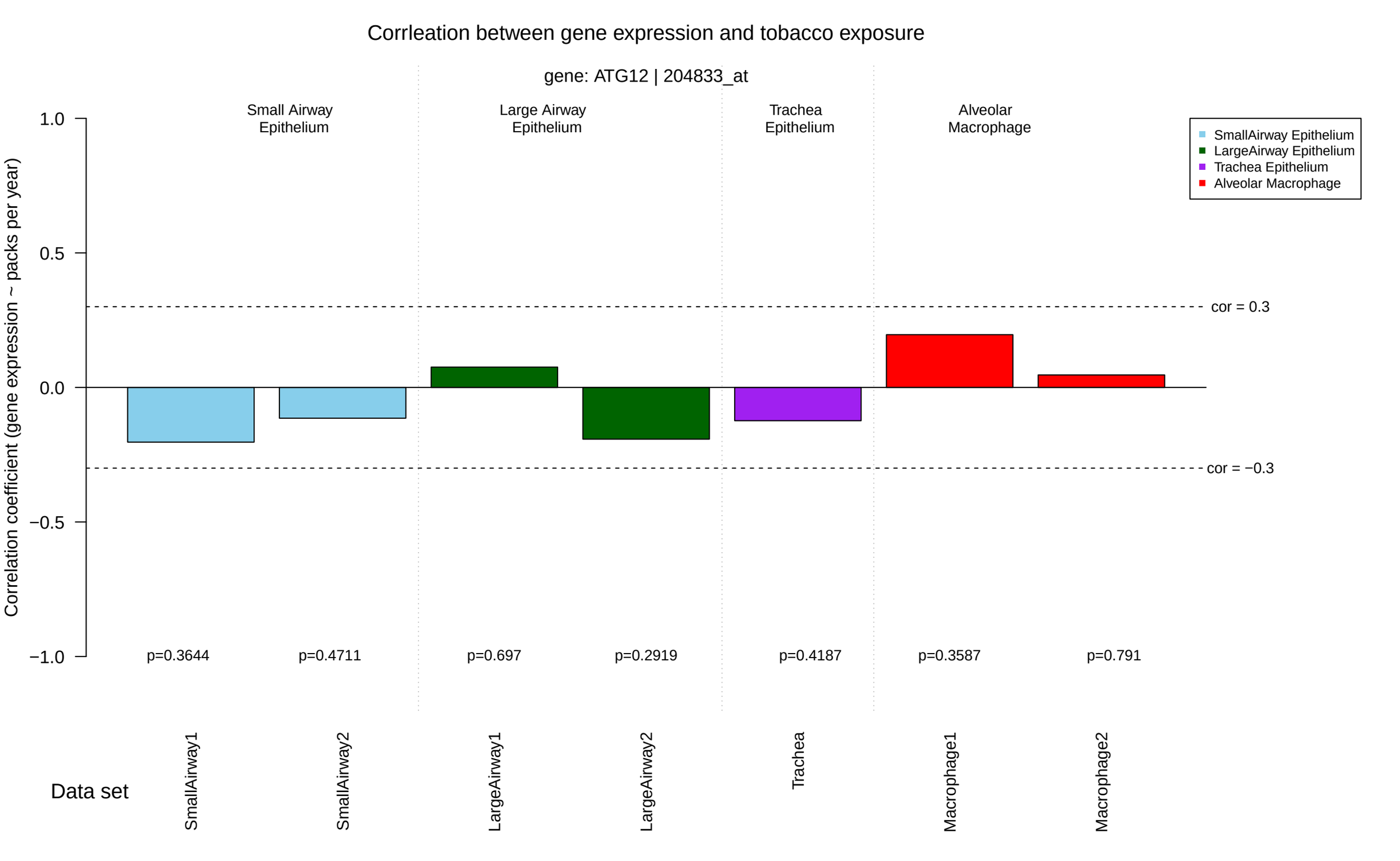

set: LargeAirway1 (p=0.3894)
ATG12 | 204833_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.247)
ATG12 | 204833_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0691)
ATG12 | 204833_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0523)
ATG12 | 204833_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4512)
ATG12 | 204833_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATG12 | 204833_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3644
p=0.4711
p=0.697
p=0.2919
p=0.4187
p=0.3587
p=0.791
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2